

Table 2 Performance comparison under disentangled local settings for Egocentric Direction and Egocentric Distance. L-Video and L-Image denote Local-Video and Local-Image settings, respectively. Values in parentheses denote absolute gains over the Global setting.

(a) Egocentric Direction

Method	Global	L-Video	L-Image
<i>Proprietary Models</i>			
Gemini-2.5-Pro	23.08	41.62 (+18.54)	66.49 (+43.41)
Gemini-3-Pro	22.11	46.38 (+24.27)	61.20 (+39.09)
GPT-5	31.09	60.32 (+29.23)	65.61 (+34.52)
GPT-4o	13.46	13.33 (-0.13)	7.05 (-6.41)
<i>Open-Source Models</i>			
Qwen3-VL-2B	16.19	11.29 (-4.90)	11.64 (-4.55)
Qwen3-VL-8B	12.50	2.82 (-9.68)	13.40 (+0.90)
InternVL3.5-2B	18.11	32.80 (+14.69)	25.40 (+7.29)
InternVL3.5-8B	16.67	12.35 (-4.32)	8.64 (-8.03)

(b) Egocentric Distance

Method	Global	L-Video	L-Image
<i>Proprietary Models</i>			
Gemini-2.5-Pro	35.40	46.94 (+11.54)	53.39 (+17.99)
Gemini-3-Pro	27.39	48.58 (+21.19)	55.79 (+28.40)
GPT-5	27.39	26.01 (-1.38)	31.04 (+3.65)
GPT-4o	10.13	23.66 (+13.53)	22.68 (+12.55)
<i>Open-Source Models</i>			
Qwen3-VL-2B	6.55	17.98 (+11.43)	14.48 (+7.93)
Qwen3-VL-8B	11.59	17.10 (+5.51)	26.12 (+14.53)
InternVL3.5-2B	4.56	7.32 (+2.76)	7.92 (+3.36)
InternVL3.5-8B	8.36	28.42 (+20.06)	25.79 (+17.43)